Name: Ruchith Chippari
Sr. Data Engineer
Email: ruchithchippari1997@gmail.com
Phone: 850-898-1610

<u>Professional Summary:</u>

- 10+ years of extensive experience in Data Engineering with strong expertise in Data Ingestion, Modelling, Querying, Processing, Analysis, and implementation of enterprise-level systems.
- Hands-on experience in the entire project life cycle including data infrastructure building, Cleansing, Manipulation, Engineering, Modelling, Optimization (performance tuning), QA, and data pipeline (ETL) to Data warehouse Deployment On-premise and on Cloud (Azure, AWS).
- Well conversant with software data analytics methodologies including Data Migration, Data Transformation, Data Extraction, Data Strategy, Data planning, Data scenarios, Test cases, and Data Analytical Reports for manual and automation of Data Transformation.
- Experience in statistical programming languages like Python, SAS, Apache Spark, and PySpark.
- Experience in analyzing data using Python, R, SQL, Microsoft Excel, Hive, PySpark, and Spark SQL for Data Mining, Data Cleansing, and Machine Learning.
- Experience in developing predictive models on large-scale datasets to address various business problems through leveraging advanced statistical modeling, machine learning, and deep learning.
- Good experience in synthesizing Data Engineering, Machine Learning, and Big Data technologies into integrated solutions.
- created Python functions for AWS Lambda, which uses Python scripts to run various analytics and transformations on massive data sets in EMR clusters.
- Experience in the development and design of various scalable systems using Hadoop technologies in multiple environments using Hive, Spark, HDFS, Map Reduce, YARN, Kafka, Pig, Hive, Sqoop, HBase, Oozie, Zookeeper, Cloudera Manager, and Horton Works.
- Experience in AWS services like VPC, Cloud front, EC2, ECS, EKS, Elastic Beanstalk, Lambda, S3, Storage gateway, RDBS, DynamoDB, Redshift, Elastic Cache, DMS, SMS, Data Pipeline, IAM, WAF, Artifacts, API gateway, SNS, SQS, SES, Auto Scaling, Cloud Formation, Cloud Watch, and Cloud Trail.
- Partnered with cross functional teams across the organization to gather requirements, architect, and develop proof of concept for the enterprise Data Lake environments like MapR, Cloudera, Hortonworks, and AWS.
- Experience working with systems processing event-based logs, user logs, machines, Azure AD, Azure Data Factory, event hub, event Queue, etc.
- Experience in building the Orchestration on Azure Data Factory for scheduling purposes.
- Strong experience writing, troubleshooting, and optimizing Spark scripts using Python, and Scala.
- Good experience with Python Libraries (Pandas, Scikit-learn, TensorFlow, Pytorch, Matplotlib, etc.)
- Hands-on experience building PySpark, Spark Java, and Scala applications for batch and stream processing involving Transformations, Actions, and Spark SQL queries on RDDs, Data frames, and Datasets.
- Experienced in using Kafka as a distributed publisher-subscriber messaging system.
- Strong experience in performance tuning of Hive queries and troubleshooting various issues related to Joins, and memory exceptions in Hive.
- Good understanding of partitioning and bucketing concepts in Hive and designed both Managed and External tables in Hive.
- Expert in data engineering and building ETL pipelines on batch and streaming data using PySpark, Spark SQL
- Experience in building and optimizing Big Data pipelines, architectures, and data sets on building Data Lakes.
- Hands-on experience in importing and exporting data between HDFS and Relational Databases using Sqoop.
- Experience in real time analytics with Spark Streaming, Kafka, and implementation of batch processing using Hadoop, Map Reduce, Pig, and Hive.
- Experienced in building highly scalable Big-data solutions using NoSQL column-oriented databases like Cassandra, MongoDB, and HBase by integrating them with Hadoop Cluster.
- Extensive experience working on ETL processes consisting of data transformation, data sourcing, mapping, conversion, and loading data from heterogeneous systems like Flat Files, Excel, Oracle, Teradata, and MS SQL Server.
- Proficient at writing MapReduce jobs and UDFs to gather, analyze, transform, and deliver the data as per business requirements and optimize the existing algorithms for best results.

- Experience working with Data warehousing concepts like Star Schema, Snowflake Schema, DataMart's, and Kimball Methodology used in Relational and Multidimensional Data Modelling.
- Experienced with version control systems like Git, GitHub, CVS, and SVN to keep the versions and configurations of the code organized.
- Proficiency in developing and refining AWS data structures, data sets, and pipelines.
- Strong experience leveraging different formats like Avro, ORC, Parquet, JSON, and Flat files.
- Sound experience in Normalization and De-normalization techniques on OLAP and OLTP systems.
- Experience with Jira, Confluence, and Rally for project management and Oozie, Airflow Scheduling Tools.
- Experienced in scripting skills in Python, Scala, and UNIX Shell.
- Experience in building interactive dashboards, performing ad-hoc analysis, and generating reports and visualizations using Tableau and PowerBI.
- Expert in importing and exporting data from different Relational Database Systems like MySQL and Oracle into HDFS and Hive using Sqoop.
- Experienced in writing SQL Queries, Stored procedures, functions, packages, tables, views, and triggers using relational databases like Oracle, DB2, MySQL, Sybase, PostgreSQL, and MS SQL server.
- Good experience working in Agile/Scrum methodologies, communication with Scrum calls for project analysis and development aspects.
- Experience in possessing extensive mathematical, and analytical skills with strong attention to detailed analysis, making significant contributions both individually and working as a team.

Technical Skills:

Programming Languages	Python, Scala, SQL, Java, C/C++, Shell Scripting
Python Libraries	Pandas, Scikit-Learn, Tensorflow, Pytorch, Matplotlib
Hadoop Technologies	Hive, Spark, HDFS, Map Reduce, YARN, Kafka, Pig, Hive, Sqoop, HBase, Oozie, Zookeeper, Cloudera Manager and Horton Works
Cloud Platforms	Azure Cloud, AWS Cloud architecture, AWS Big Data computing
Machine Learning	Decision Tree, SVM, Naïve Bayes, Regression, Classification, Neutral networks
Databases	Oracle, DB2, MySQL, Sybase, PostgreSQL. MS SQL Server
NoSQL Databases	MongoDB, Cassandra, HBase
Visualization Tools	Tableau, Power BI, Advanced Excel
Version Control Systems	Git, GitHub, CVS, SVN
IDEs	PyCharm, IntelliJ IDEA, Jupyter Notebooks, Eclipse
Operating Systems	Unix, Linux, Windows

Professional Experience:

Client: Kroger - Columbus, Ohio **December 2022 - Present**

Sr. Data Engineer

Responsibilities:

- Participated in requirement grooming meetings which involves understanding functional requirements from a business perspective and providing estimates to convert those requirements into software solutions.
- Performed analysis on existing data flows and created high level/low level technical design documents for business stakeholders that confirmed technical design aligns with business requirements.
- Used Pandas, Scikit-Learn, Tensorflow, Pytorch, and Matplotlib in Python for developing data pipelines and various machine learning algorithms.
- Create and implement ETL procedures in AWS Glue to move data into AWS Redshift from external sources such as S3 and Parquet/Text files.
- Used AWS EMR to move large data (Big Data) into other platforms such as AWS data stores, Amazon S3, and Amazon Dynamo DB.
- Worked on the creation and deployment of Spark jobs in different environments and loading data to NoSQL database Cassandra/Hive/HDFS. Secure the data by implementing encryption.
- Developed code using: Apache Spark and Scala, IntelliJ, NoSQL databases (Cassandra), Jenkins, Docker pipelines, GITHUB, Kubernetes, HDFS file System, Hive, Kafka for streaming Real time streaming data, Kibana for monitor logs, etc. authentication/authorization to the data.

- Large volumes of data were transformed and moved into and out of different AWS databases and data stores, including Amazon DynamoDB and Amazon Simple Storage Service (Amazon S3), using AWS EMR.
- Extracted, Transformed, and Loaded data from source systems to Azure Data Storage services using a combination of Azure Data Factory, T-SQL, Spark SQL, and U-SQL Azure Data Lake Analytics.
- Worked on Data Ingestion to one or more Azure Services - (Azure Data Lake, Azure Storage, Azure SQL, and Azure DW) and processing the data in Azure Databricks.
- Ingested API data into Azure Data Storage services (Blob) using EventHub and PySpark in Databricks. Data was further transformed into a structured format and stored in DataLake gen2.
- Developed the PySpark code for AWS Glue jobs and for EMR.
- Implemented business logic to perform complex data operations, created Dim and Fact tables in Data warehouse using Data factory as an orchestration tool, developed a layer SSAS cube which in-turn was used by PowerBI/Tableau for reporting dashboard services, created automated logic app workflows to send out daily reports.
- Worked on PySpark and Spark SQL transformation in Azure Databricks to perform complex transformations for business rule implementation
- utilized AWS Lambda to automate the process and AWS Glue for transformations.
- Migrated data pipelines, from Azure to Snowflake multi-cluster data warehouse as it handles semi-structured data which are in turn connected to analytical engines for reporting.
- Built deployment automation ETL CI/CD pipelines for deployment of API solutions and database changes using, Azure DevOps maintained codebase in Azure git, wrote unit/regression/integration tests for successful production deployment and post-deployment validations.
- Created several Databricks Spark jobs with PySpark to perform several tables to table operations
- Used Apache Airflow to build data pipelines and used various airflow operators like bash operator Hadoop operator and branching operator.
- Creating many AWS Lambda functions and API Gateways for data submission via Lambda function.
- Configured and implemented the Azure Data Factory Triggers and scheduled the Pipelines. Created and configured a new event hub with the provided event hubs namespace.
- Responsible for deployments to DEV, QA, PRE-PROD (CERT), and PROD using Azure.
- Responsible for facilitating load data pipelines and benchmarking the developed product with the set performance standards.
- Developed and produced a dashboard, and key performance indicators and monitored organization performance.
- Defined data needs, evaluated data quality, and extracted/transformed data for analytic projects and research.
- Tested the database using complex SQL scripts and handled the performance issues effectively.
- Worked with business users to flush out requirements, envision what data (KPI) is required and how it will look after its pumped through Tableau & Power BI tools from Amazon kinesis on a real-time basis.

Environment: Python, Pandas, Scikit-Learn, Tensorflow, Pytorch, Matplotlib, Azure Data Lake, Azure Storage, Azure Data Factory, Azure SQL, Azure DW, Azure DevOps, Databricks, Snowflake Schema, Machine Learning, ETL, Hadoop, Spark, PySpark, Kafka, REST API, PySpark, SQL, Power BI, Apache Airflow, Jenkins, Power BI, Tableau

Client: Barclays - New York City, New York **November 2020 - November 2022**
Sr. Data Engineer
Responsibilities:

- Built machine learning predictive ETL pipelines which were accessed using REST API built out of geological data, interpreting seismic data, and optimizing the oil exploration process.
- Developed Spark applications using PySpark and Spark-SQL for data extraction, transformation, and aggregation from multiple file formats for analyzing and transforming the data to uncover insights into customer usage patterns.
- Built environment and deployment automation, infrastructure-as-code, deployment data pipeline specification and development using utilities like GitHub Actions using Terraform, Azure DevOps, and CI/CD automated pipelines for release cycles.
- Built Apache Airflow with AWS to analyze multi-stage machine learning processes with Amazon Sage Maker tasks.
- utilized S3, EC2, AWS Glue, Athena, RedShift, EMR, SNS, SQL, DMS, and Kenesis, among other AWS services.
- Implemented Azure Logic Apps, Azure Functions, Azure Storage, and Service Bus queries for large enterprise-level ERP integration systems.
- Built complex ETL jobs that transform data visually with data flows or by using compute services Azure Databricks, and Azure SQL Database

- Worked in Azure environment for development and deployment of Custom Hadoop Applications.
- Designed and implemented scalable Cloud Data and Analytical architecture solutions for various public and private cloud platforms using Azure.
- Involved in ingestion, transformation, manipulation, and computation of data using kinesis, SQL, AWS glue and Spark
- Implemented various Azure platforms such as Azure SQL Database, Azure SQL Data Warehouse, Azure Analysis Services, HDInsight, Azure Data Lake, and Data Factory.
- Extracted and loaded data into the Data Lake environment (MS Azure) by using Sqoop which was accessed by business users.
- Create data ingestion modules using AWS Glue for loading data in various layers in S3 and reporting using Athena and Quicksight.
- Migrated data warehouses to Snowflake Data warehouse. Defined virtual warehouse sizing for Snowflake for different types of workloads.
- Extracted data from Data Lakes, and EDW to relational databases for analysing and getting more meaningful insights using SQL Queries and PySpark.
- Experienced with event-driven and scheduled AWS Lambda functions to trigger various AWS resources.
- Worked on Spark improving the performance and optimization of the existing algorithms in Hadoop using Spark Context, Spark-SQL, Data Frame, Pair RDD's
- Utilized Spark-SQL & PySpark based models on Azure Databricks ML workspace, to build & deploy ML models at scale for suspicious phishing, COVID-19 impact, customer activity models, ecosystem, retention/stickiness, etc.
- Developed PySpark script to merge static and dynamic files and cleanse the data.
- Designed, developed, and did maintenance of data integration programs in a Hadoop and RDBMS environment with both traditional and non-traditional source systems.
- Created PySpark procedures, functions, and packages to load data.
- Developed MapReduce programs to parse the raw data, populate staging tables, and store the refined data in partitioned tables in the EDW.
- Worked on monitoring and troubleshooting the Kafka-storm-HDFS data pipeline for real time data ingestion in Data Lake in HDFS. Wrote Sqoop Scripts for importing and exporting data from RDBMS to HDFS.
- Solved performance issues in Hive and Pig scripts with an understanding of Joins, Group, and Aggregation and how they translate to MapReduce jobs.
- Designed workflows using Airflow to automate the services developed for change data capture.
- Carried out data transformation and cleansing using SQL queries and PySpark.
- Used Kafka and Spark Streaming to ingest real time or near real time data in HDFS.
- Worked on PySpark APIs for data transformations.
- Created dashboards as part of Data Visualization using Power BI.

Environment: Python, Azure Cloud (Databricks, Data Factory, Cosmos DB, Function App, Data Lake Gen2, Logic Apps, Azure Storage, Service Bus, Azure SQL), Machine Learning, Hadoop, Spark, Spark SQL, PySpark, Hive, HDFS, Pig, Sqoop, MapReduce, Apache Kafka, Airflow, Snowflake, SQL DB, SQL, Power BI, Git

Client: Walgreens, Deerfield, Illinois **January 2019 - October 2020**
Role: Data Engineer
Responsibilities:

- Built high-quality Data warehouses and data lakes at an enterprise level, worked with cross-functional teams to automate data ingestion and schedule jobs at daily, weekly & monthly frequencies on AWS cloud.
- Modernized the data warehouse environment by using AWS cloud-based Hadoop system, data manipulations, and maintained tables for reporting, data science, dashboarding, and ad-hoc analyses.
- Utilized Python, Spark on AWS Elastic Search, and Maker to develop & execute Analytics & Machine learning models for fraud detection and risks.
- Developed ETL Processes in AWS Glue to migrate data from external sources like S3, ORC/Parquet/Text Files into AWS Redshift.
- Created AWS Lambda functions and assigned IAM roles to schedule Python scripts using Cloud Watch Triggers to support the infrastructure needs (SQS, Event Bridge, SNS)
- Built logical and physical data models for Snowflake for different types of workloads and developed large scale data intelligence solutions around the data warehouse.

- Ingested data by going through cleansing and transformations and leveraging AWS Lambda, AWS Glue, and Step Functions.
- Used Python for pattern matching in build logs to format warnings and errors.
- Designed and deployed Hadoop cluster and different big data analytic tools including Pig, Hive, HBase, and Sqoop.
- Developed simple and complex MapReduce programs in Hive, Pig, and Python for Data Analysis on different data formats.
- Wrote Spark applications using Scala to interact with the database using Spark SQL Context and accessed Hive tables using Hive Context.
- Involved in designing different components of the system like big-data event processing framework Spark, distributed messaging system Kafka, and SQL database.
- Used PySpark data frame to read text data, CSV data, and image data from HDFS, S3, and Hive.
- Managed and scheduled Jobs on Hadoop Cluster using UC4 (Confidential Preoperatory Scheduling Tool) workflows. Managed and reviewed Hadoop and HBase log files.
- Worked on AWS POC for transferring data from local file system to S3.
- Imported semi-structured data from Avro files using Pig to make serialization faster
- Managed large datasets using Panda data frames and MySQL.
- Implemented Spark Streaming and Spark SQL using Data Frames.
- Created multiple Hive tables, and implemented Dynamic Partitioning and Buckets in Hive for efficient data access.
- Involved in converting MapReduce programs into Spark transformations using Spark RDD on Python.
- Implemented data quality, integrity, and reliability and created spark performance tuning solutions through the data pipeline by designing, maintaining, and promoting data governance, enrichment & database efficiency.
- Extracted data from medical devices stored by scientists in different network drives, loaded into NOSQL DB (Mongo) to expedite research studies from internal & third-party sources for the R&D department through the use of algorithms.
- Involved in building unit tests and integration tests for automated data validation quality checks before deployment.
- Built the analytics data warehouse and cubes on cloud-based Hadoop, Git version control system, automatic deployment using Jenkins & scheduling system.
- Created several complex Reports (sub reports, graphical, multiple groupings, drill-downs, parameter driven, formulas, summarized and conditional formatting) in Power BI for visualization of data.

Environment: Python, AWS (S3, Elastic Search, Sage maker, Glue, Redshift, SQS, SNS, Lambda, Step Functions, Event Bridge, CloudWatch, IAM), Machine Learning, Snowflake, Hadoop, Spark, Spark SQL, Scala, HBase, Hive, Pig, Sqoop, MapReduce, HDFS, Kafka, MongoDB, Git, Jenkins, Power BI

Client: AT&T, Atlanta GA **January 2017 - December 2018**
Role: Hadoop Data Engineer
Responsibilities:

- Responsible for continuous monitoring and managing the Hadoop cluster through Hortonworks (HDP) distribution.
- Worked on Spark RDD, Data Frame API, Data set API, Data Source API, Spark SQL, and Spark Streaming.
- Developed Spark Applications by using Python and implemented Apache Spark data processing Project to handle data from various RDBMS and Streaming sources.
- Worked with Spark to improve the performance and optimization of the existing algorithms in Hadoop.
- Used Spark Context, Spark-SQL, Spark MLLib, Data Frame, Pair RDD, and Spark YARN.
- Used Spark Streaming APIs to perform transformations and actions on the fly for building common.
- Developed Kafka consumer API in Python for consuming data from Kafka topics.
- Consumed Extensible Markup Language (XML) messages using Kafka and processed the XML file using Spark Streaming to capture User Interface (UI) updates.
- Developed Pre-processing job using Spark Data frames to flatten JSON documents to flat file.
- Wrote live Real-time Processing and core jobs using Spark Streaming with Kafka as a Data pipeline system.
- Worked on AWS Data Pipeline to configure data loads from S3 to Redshift and have used AWS components (Amazon Web Services) - Downloading and uploading data files (with ETL) to AWS system using S3 components.
- Used AWS Data Pipeline to schedule an Amazon EMR cluster to clean and process web server logs stored in an Amazon S3 bucket.
- Developed Lambda functions and assigned IAM roles to run Python scripts along with various triggers (SQS, SNS).
- Developed and deployed AWS Lambda services for ETL migration services by generating a serverless data pipeline that can be written to Glue and queried from Athena.

- Stored incoming data in the Snowflakes staging area. Created numerous ODI interfaces and load into Snowflake DB.
- Designed columnar families in Cassandra and ingested data from RDBMS, performed data transformations, and then exported the transformed data to Cassandra as per the business requirement.
- Experienced in creating data models for client's transactional logs, and analyzing the data from Cassandra.
- Used Hive QL to analyze the partitioned and bucketed data, Executed Hive queries on Parquet tables.
- Stored in Hive to perform data analysis to meet the business specification logic.
- Developed Custom UDF in Python and used UDFs for sorting and preparing the data.
- Worked on Custom Loaders and Storage Classes in PIG to work on several data formats like JSON, XML, CSV, and generated Bags for processing using Pig, etc.
- Developed Sqoop and Kafka Jobs to load data from RDBMS, and External Systems into HDFS and HIVE.
- Developed Oozie coordinators to schedule Hive scripts to create Data pipelines.
- Wrote several Map Reduce Jobs using PySpark, and NumPy and used Jenkins for Continuous integration.
- Worked on cluster and testing of HDFS, Hive, Pig, and MapReduce to access cluster for new users.
- Responsible for continuous monitoring and managing the Hadoop cluster through Cloudera Manager.
- Maintained Tableau functional reports based on user requirements.

Environment: Python, Amazon Web Services (AWS), Spark, Spark-Streaming, Spark SQL, MapReduce, HDFS, Hive, Pig, Apache Kafka, Sqoop, Python, PySpark, Shell scripting, Linux, MySQL, Oracle Enterprise DB, Jenkins, Eclipse, Oracle, Git, Oozie, Tableau, MySQL, Soap, XML, Cassandra, Agile Methodologies

Client: Larsen & Toubro Infotech (LTI) - Mumbai, India **July 2013 - October 2016**
Role: Big Data Engineer
Responsibilities:

- Worked with Hadoop ecosystem components like HBase, Sqoop, Zookeeper, Oozie, Hive, and Pig with Cloudera Hadoop distribution.
- Involved in start to end process of Hadoop jobs that used various technologies such as Sqoop, Pig, Hive, MapReduce, Spark, and Shells scripts.
- Managed and supported enterprise Data Warehouse operation and Big Data application development using Cloudera and Hortonworks HDP.
- Developed Pig scripts to transform the raw data into intelligent data as specified by business users.
- Installed Hadoop, Map Reduce, and HDFS to develop multiple MapReduce jobs in Pig and Hive for data cleansing and pre-processing.
- Used Spark API over Hortonworks Hadoop YARN to perform analytics on data in Hive.
- Improved the performance and optimization of the existing algorithms in Hadoop using Spark Context, Spark-SQL, Data Frame, Pair RDDs, and Spark YARN.
- Developed Spark code using Scala and Spark-SQL/Streaming for faster testing and processing of data.
- Developed a Spark job in Java that indexes data into Elastic Search from external Hive tables which are in HDFS.
- Performed transformations, cleaning, and filtering on imported data using Hive, and MapReduce, and loaded final data into HDFS.
- Explored with Spark improving the performance and optimization of the existing algorithms in Hadoop using Spark Context, Spark-SQL, Data Frame, Pair RDDs, and Spark YARN.
- Import the data from different sources like HDFS/HBase into Spark RDD and develop a data pipeline using Kafka and Storm to store data into HDFS.
- Used Spark Streaming to receive real time data from Kafka and store the stream data to HDFS using Scala and NoSQL databases such as HBase and Cassandra.
- Documented the requirements including the available code which should be implemented using Spark, Hive, HDFS, and HBase.
- Utilized PySpark to distribute data processing on large streaming datasets to improve ingestion and process.
- Performed transformations like event joins, filter boot traffic, and some pre-aggregations using Pig.
- Executed Hive queries on Parquet tables stored in Hive to perform data analysis to meet the business requirements.
- Configured Oozie workflow to run multiple Hive and Pig jobs which run independently with time and data availability.
- Imported and exported the analyzed data to the relational databases using Sqoop for visualization and to generate reports for the BI team.
- Created several types of data visualizations using Python and Tableau.

Environment: Cloudera, Hortonworks, Hadoop, Sqoop, Pig, Hive, MapReduce, Spark, PySpark, Shell Scripting, SQL, Hortonworks, Python, Java, Scala, MLLib, HDFS, YARN, Kafka, Zookeeper, Oozie, Cassandra, Tableau, Agile